



FORT STREET HIGH SCHOOL

2008

HIGHER SCHOOL CERTIFICATE COURSE

Name: _____

Teacher: _____

Class: _____

Assessment Task 3

Mathematics Extension I

Time allowed: 1 hour

Outcomes Assessed	Questions
Determines integrals by reduction to a standard form through a given substitution.	1
Uses the relationship between functions, inverse functions, their derivatives and integrals.	2, 3
Synthesises mathematical solutions to harder problems and communicates them in appropriate form.	4, 5

Question	1	2	3	4	5	Total	%
Marks	/11	/11	/11	/11	/11	/55	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started on a new page

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Total marks – 55

Attempt Questions 1 – 5

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1. (11 marks)

a) Find $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ using the substitution $u = e^x$ **3**

b) Evaluate $\int_3^4 x\sqrt{4-x} dx$ using the substitution $x = 4-u$ **3**

c) i) Express $\cos 2x$ in terms of $\cos x$. **1**

ii) Hence using the substitution $x = \sin^2 \theta$, evaluate $\int_0^1 \sqrt{\frac{1-x}{x}} dx$ **4**

Question 2. (11 marks)

- a) i) A function is defined by $f(x) = |x - 2|$.
Explain why $f(x)$ does not have an inverse function. **1**
- ii) State the largest positive domain which will define the inverse function $f^{-1}(x)$. **1**
- iii) Find $f^{-1}(x)$ in terms of x and state its domain and range. **3**
- b) i) State the domain and range of $f(x) = \frac{1}{2} \cos^{-1}(x - 1)$. **2**
- ii) Hence sketch the graph of $f(x)$ clearly showing coordinates of end points. **2**
- c) Find the exact value of $\sin^{-1} \frac{\sqrt{3}}{2} + \cos^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} \left(\sin \frac{5\pi}{6} \right)$ **2**

Question 3. (11 marks)

- | | | | | |
|----|--|------|---|----------|
| a) | Differentiate | i) | $\sin^{-1}\left(\frac{3x}{2}\right)$ | 2 |
| | | ii) | $\cos^{-1}(5x)$ | 2 |
| | | iii) | $\tan^{-1}\left(\tan\frac{x}{2}\right)$ | 3 |
| b) | Find the primitive of $\frac{1}{9+4x^2}$ | | | 2 |
| c) | Find $\int (4-x^2)^{-\frac{1}{2}} dx$ | | | 2 |

Question 4. (11 marks)

a) i) Write $x^2 + 4x + 5$ in the form $(x + a)^2 + b^2$ **1**

ii) Hence evaluate $\int_{-3}^{-1} \frac{dx}{x^2 + 4x + 5}$ **3**

b) i) Differentiate $\sin^{-1} x + \sqrt{1 - x^2}$ **3**

ii) Hence evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{1-x}{1+x}} dx$ **4**

Question 5. (11 marks)

a) Evaluate $\int_0^3 \frac{1}{3+x^2} dx$ **3**

b) i) Show the curves $y = \cos^{-1} x$ and $y = 2 \tan^{-1}(1-x)$ intersect the y axis at the same point. **3**

ii) Hence, or otherwise, show there is a common tangent at this point. **2**

c) A region is bounded by the curve $y = \sin^{-1} x$, the x - axis and the line $x = 1$. **3**

Use Simpson's rule with 3 function values to find the approximation to the area of the region correct to two decimal places.

Question 1 (11 marks)

a) $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$, $u = e^x$
 $du = e^x dx$ ✓

$$\int e^x (1-e^{2x})^{-\frac{1}{2}} dx$$
 ✓

$$\int \frac{du}{\sqrt{1-u^2}}$$

$$\sin^{-1} u + c$$

$$\sin^{-1} e^x + c$$
 ✓

b) $\int_3^4 x \sqrt{4-x} dx$

$x = 4-u$, $u = 4-x$
 $dx = -du$
 if $x=3$, $u=1$ ✓
 $x=4$, $u=0$

$$-\int_1^0 (4-u) u^{\frac{1}{2}} du$$

$$\int_0^1 4u^{\frac{1}{2}} - u^{\frac{3}{2}} du$$
 ✓

$$\left[\frac{2 \cdot 4 u^{\frac{3}{2}}}{3} - \frac{2u^{\frac{5}{2}}}{5} \right]_0^1$$

$$\frac{2 \cdot 4 \cdot 1^{\frac{3}{2}}}{3} - \frac{2 \cdot 1^{\frac{5}{2}}}{5} - (0-0)$$

$$\frac{8}{3} - \frac{2}{5}$$

$$\frac{34}{15}$$
 ✓

Generally well done

Some students did not find limits in terms of u .

c) i) $\cos 2x = 2\cos^2 x - 1$ ① ✓

ii) $x = \sin^2 \theta$

$$dx = 2\sin \theta \cos \theta d\theta$$

if $x=0$, $\sin^2 \theta = 0$
 $\theta = 0$

$x=1$, $\sin^2 \theta = 1$ ✓
 $\theta = \frac{\pi}{2}$

$$\int_0^1 \sqrt{\frac{1-x}{x}} dx = \int_0^{\frac{\pi}{2}} \sqrt{\frac{1-\sin^2 \theta}{\sin^2 \theta}} \cdot 2\sin \theta \cos \theta d\theta$$
 ✓

$$= \int_0^{\frac{\pi}{2}} \frac{\cos^2 \theta}{\sin^2 \theta} \cdot 2\sin \theta \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta} \cdot 2\sin \theta \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{2}} 2\cos^2 \theta d\theta$$
 ✓

$$= \int_0^{\frac{\pi}{2}} (\cos 2\theta + 1) d\theta$$

$$= \frac{\sin 2\theta}{2} + \theta \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\sin \pi}{2} + \frac{\pi}{2} - \left(\frac{\sin 0}{2} + 0 \right)$$
 ✓

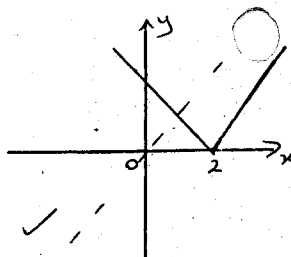
$$= \frac{\pi}{2}$$

Some students did not find the relations between dx and $d\theta$.

Question 2 (11 marks).

a) i) $f(x) = |x-2|$

$f(x)$ does not have an inverse function because it cannot be reflected in the line $y=x$ to give one to one correspondence.



Some students did not clearly explain why the inverse doesn't exist.

ii) $x \geq 2$ will define the function

iii) $y = |x-2|$

$x = y-2$

$x+2 = y$

$\therefore f^{-1}(x) = x+2$

Domain: $\{x: x \geq 0\}$

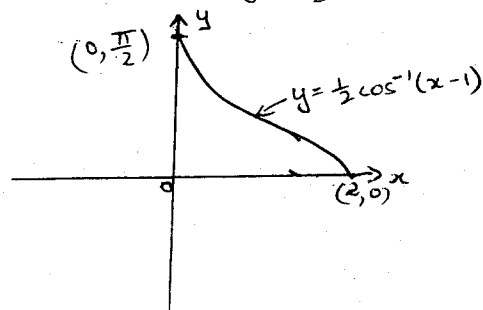
Range: $\{y: y \geq 2\}$

b) i) $f(x) = \frac{1}{2} \cos^{-1}(x-1)$

Domain: $-1 \leq x-1 \leq 1$
 $0 \leq x \leq 2$

Range: $0 \leq y \leq \pi$

$0 \leq \frac{1}{2}y \leq \frac{\pi}{2}$



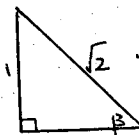
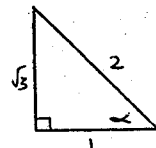
Many students did not know how to interchange the x and y to get $f^{-1}(x)$.

$y \neq -x+2$ as $x \geq 2$.

Good but Domain & range need to be clearly stated which is which.

c) $\sin^{-1} \frac{\sqrt{3}}{2} \bigcirc \cos^{-1} \frac{1}{\sqrt{2}} = \sin^{-1}(\sin \frac{5\pi}{6})$

Let $\alpha = \sin^{-1} \frac{\sqrt{3}}{2}$, $\beta = \cos^{-1} \frac{1}{\sqrt{2}}$



$\therefore \alpha = \frac{\pi}{3}$

$\beta = \frac{\pi}{4}$

and $\sin^{-1}(\sin \frac{5\pi}{6}) = \frac{\pi}{6}$

so $\sin^{-1} \frac{\sqrt{3}}{2} + \cos^{-1} \frac{1}{\sqrt{2}} = \sin^{-1}(\sin \frac{5\pi}{6})$

$\frac{\pi}{3} + \frac{\pi}{4} = \frac{\pi}{6}$

$\frac{4\pi + 3\pi - 2\pi}{12}$

$\frac{5\pi}{12}$

Evaluation of $\sin^{-1}(\sin \frac{5\pi}{6})$

caused problems for some students

$\sin^{-1}(\sin \frac{5\pi}{6}) \neq \frac{5\pi}{6}$

since

$-\frac{\pi}{2} < y < \frac{\pi}{2}$

Question 3 (11 marks)

$$a) i) \frac{d}{dx} \sin^{-1}\left(\frac{3x}{2}\right) = \frac{3}{\sqrt{2^2 - (3x)^2}} \checkmark$$

$$= \frac{3}{\sqrt{4 - 9x^2}} \checkmark$$

$$ii) \frac{d}{dx} \cos^{-1} 5x = \frac{-1}{\sqrt{1 - (5x)^2}} \times 5 \checkmark$$

$$= \frac{-5}{\sqrt{1 - 25x^2}} \checkmark$$

$$iii) \frac{d}{dx} \tan^{-1}\left(\tan \frac{x}{2}\right) \quad , \quad \text{let } u = \tan \frac{x}{2}$$

$$du = \frac{1}{2} \sec^2 \frac{x}{2} \cdot \frac{1}{2}$$

$$\frac{1}{1 + \left(\tan \frac{x}{2}\right)^2} \times \frac{1}{2} \sec^2 \frac{x}{2}$$

$$\frac{1}{\sec^2 \frac{x}{2}} \times \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} \checkmark$$

Alternate method

$$\frac{d}{dx} \tan^{-1}\left(\tan \frac{x}{2}\right) \checkmark$$

$$\frac{d}{dx} \frac{x}{2} \checkmark \text{ since } \tan^{-1}(\tan x) = x$$

$$\frac{1}{2} \checkmark$$

$$b) f'(x) = \frac{1}{9 + 4x^2}$$

$$f(x) = \frac{1}{4} \int \frac{1}{\frac{9}{4} + x^2} dx \checkmark$$

$$= \frac{1}{4} \times \frac{2}{3} \int \frac{\frac{3}{2}}{\left(\frac{3}{2}\right)^2 + x^2} dx = \frac{1}{6} \tan^{-1} \frac{2x}{3} + c \checkmark$$

$$c) \int (4 - x^2)^{-\frac{1}{2}} dx = \int \frac{1}{\sqrt{4 - x^2}} dx \checkmark$$

$$= \sin^{-1} \frac{x}{2} + c \checkmark$$

Incorrect manipulation of a^2 and x^2

Mostly well done

Many students did not correctly differentiate $\tan \frac{x}{2}$.

Others failed to recognise that $1 + \left(\tan \frac{x}{2}\right)^2 = \sec^2 \frac{x}{2}$

$\tan^{-1}(\tan x) = x$

Many students had difficulty finding correct co-eff. of $\tan^{-1} \frac{2x}{3}$

Well done by those students who recognised $(4 - x^2)^{-\frac{1}{2}} = \frac{1}{\sqrt{4 - x^2}}$

Question 4 (11 marks)

$$a) i) x^4 + 4x + 2^2 + 5 - 2^2$$

$$(x+2)^2 + 1 \checkmark$$

$$ii) \int_{-3}^{-1} \frac{dx}{x^2 + 4x + 5}$$

$$\int_{-3}^{-1} \frac{dx}{(x+2)^2 + 1} \checkmark$$

$$\tan^{-1}(x+2) \Big|_{-3}^{-1}$$

$$\tan^{-1}(-1+2) - \tan^{-1}(-3+2) \checkmark$$

$$\tan^{-1} 1 - \tan^{-1}(-1)$$

$$\frac{\pi}{4} + \frac{\pi}{4}$$

$$\frac{\pi}{2} \checkmark$$

$$b) i) \frac{d}{dx} (\sin^{-1} x + \sqrt{1 - x^2})$$

$$\frac{d}{dx} [\sin^{-1} x + (1 - x^2)^{\frac{1}{2}}] \checkmark$$

$$\frac{1}{\sqrt{1 - x^2}} + \frac{1}{2} (1 - x^2)^{-\frac{1}{2}} \times -2x \checkmark$$

$$\frac{1}{\sqrt{1 - x^2}} - \frac{x}{\sqrt{1 - x^2}} \checkmark$$

$$\frac{1 - x}{\sqrt{1 - x^2}}$$

$$\frac{1 - x}{\sqrt{(1 - x)(1 + x)}}$$

$$\sqrt{\frac{1 - x}{1 + x}} \checkmark$$

Usually well done

Some student had difficulty with the substitution into inverse tan. so values such as $\tan^{-1}(-3)$ were difficult work with.

Incorrect use of chain rule particularly use of minus sign

This mark was also given if the simplification occurred in ii)

ii) $\int_0^{\frac{1}{2}} \sqrt{\frac{1-x}{1+x}} dx$

$$\left[\sin^{-1} x + \sqrt{1-x^2} \right]_0^{\frac{1}{2}}$$

$$\sin^{-1}\left(\frac{1}{2}\right) + \sqrt{1-\frac{1}{4}} - \sin^{-1}0 - \sqrt{1}$$

$$\frac{\pi}{6} + \frac{\sqrt{3}}{2} - 1$$

Question 5 (11 marks)

a) $\int_0^3 \frac{1}{3+x^2} dx = \frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \Big|_0^3$

$$= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{3}{\sqrt{3}} - \tan^{-1} 0 \right]$$

$$= \frac{1}{\sqrt{3}} \times \frac{\pi}{3}$$

$$= \frac{\pi\sqrt{3}}{9}$$

b) i) $y = \cos^{-1} x$ and $y = 2 \tan^{-1}(1-x)$

At the y axis $x=0$

so $y = \frac{\pi}{2}$ and $y = 2 \times \frac{\pi}{4} = \frac{\pi}{2}$

$\therefore$ curves intersect at $(0, \frac{\pi}{2})$ on the y axis

ii) $y = \cos^{-1} x$ and $y = 2 \tan^{-1}(1-x)$

$$y' = \frac{-1}{\sqrt{1-x^2}} \quad \text{and} \quad y' = \frac{-2}{1+(1-x)^2}$$

When $x=0$, $y' = -1$

$$y' = \frac{-2}{1+1} = -1$$

$\therefore$ tangents have the same

The link between the derivative in part i) was missed because derivative was not simplified sufficiently.

"Hence" was ignored by some students.

Mostly well executed

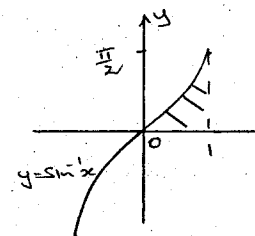
Some students did not use the table of standard integrals, so omitting $\frac{1}{\sqrt{3}}$.

Some students attempted to find x intercept instead of y.

Inaccuracies with the sign of y' impeded some students.

c)

x	0	$\frac{1}{2}$	1
$f(x)$	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$



$$A = \int_0^1 f(x) dx$$

$$= \frac{b-a}{6} \left[f(a) + f(b) + 4f\left(\frac{a+b}{2}\right) \right]$$

$$= \frac{1}{6} \left[0 + \frac{\pi}{2} + 4 \times \frac{\pi}{6} \right]$$

$$= \frac{7\pi}{36}$$

$$\approx 0.610865238$$

$$= 0.61 \text{ (to 2 d.p.)}$$

area is approx 0.61 u².

Poorly done by a no. of students

- some did not know rule

- others used incorrect no. of function values

- h incorrect

- did not use radian measure

- did not round off value correctly